

exactly 80 mm: _____

FIRST SOLVE

New here? Do the walkthrough once at [cubepath-six.vercel.app/](#) or [new here!](#) This card is the memory jog.

NOTATION

R = F, L, U = turn that face clockwise, looking at it from outside the cube.
 = counter-clockwise, 2 = half turn. Lowercase r f = that face plus the layer behind it, turning together.
 y = spin the whole cube left, white staying center. A whole-cube turn solves nothing.

WORDS

algorithm = a fixed sequence of turns - trigger = a short algorithm your hands run as one unit

1 WHITE CROSS



White center DOWN. Bring the four white edges to it. Each edge's side color must match the center it touches. No algorithm – work it out.

Can't see it? Build the four white edges on the YELLOW face first, check under its matching center, then turn each one down 180 degrees.

2 WHITE CORNERS

Find a white corner on top, turn U until it sits above the empty slot it belongs in, then match a picture below and repeat until it drops in.



white sticker RIGHT
RURU'



white sticker FRONT
L'UL'



white sticker U run the first one once to knock it out, then look again
RURU'

3 MIDDLE EDGES

Top edge with NO yellow: turn U until its front color matches the center under it. The arrow shows which slot it goes to.



goes RIGHT
U RURU' y L'UL'



goes LEFT
U' L'UL' y' RURU'

FIRST SOLVE

New! Here's Do the walkthrough once at [cubepath-six.vercel.app](#)/app/learn. This card is the memory jog.

NOTATION

R = 1/2 turn clockwise, looking at it from outside the cube
 counter-clockwise = 2 = half turn
 Lowercase r f = that face plus the layer behind it, turning together.
 y = spin the whole cube left, white staying center. A whole-cube turn solves nothing.

WORDS

algorithm = a fixed sequence of turns + trigger = a short algorithm
 your hands run as one unit

1 WHITE CROSS



White center DOWN. Bring the four white edges to it. Each edge's side color must match the center it touches. No algorithm – work it out.

Can't see it? Build the four white edges on the YELLOW face first, each above its matching center, then turn each one down 180 degrees.

2 WHITE CORNERS

Find a white corner on top, turn U until it sits above the empty slot it belongs in, then match a picture below and repeat until it drops in.



white sticker RIGHT

R' U' R



white sticker FRONT

L' U' L



white sticker UP run the first one once to knock it out, then look again

R' U' R

3 MIDDLE EDGES

Top edge with NO yellow: turn U until its front color matches the center under it. The arrow shows which slot it goes to.

goes RIGHT

U R U' R' y L' U' L



goes LEFT

U' L' U' L y' R' U' R'



TWO-LOOK		OLL CORNERS - continued	
	Line bar left-to-right FURU'U'f'		Headlights 2 2 yellow, headlights at the back R2DR'U2 RD'R'U2R'
	Hook hook rotating front-right FURU'U'f'		2x Headlights 2 0 yellow, headlights both sides GRU'U RU'R' URU2R'
	Sune 2 1 yellow, rest clockwise SUNE'U RU2R'		Chameleon 2 2 yellow, next to each other FUR'U'r' FRF'
	Anti-Sune 2 1 yellow, rest counter-clockwise RU2R' U'RU'R'		Bowtie 2 2 yellow diagonal F'URU'U'r' FR
	Pi 2 0 yellow, headlights left FURU'U'f' FRURU'U'f'	FALLBACK The badge on each row is your fallback: that many Sunes, with a U turn between, also finishes the case. Machine-checked: there is the word. Top not all yellow and nothing matches? Run Sune and read again.	

TWO-LOOK OLL CROSS -yellow edges	OLL CORNERS -continued
 Line bar left-to-right FRUR'U'f'	 Headlights $\otimes$ 2 yellow, headlights at the back R2DR'U2 RD'R'U2R'
 Hook hook pointing front-right FRUR'U'f'	 2s Headlights $\otimes$ 0 yellow, headlights both sides RU'R'U RU'R' URU2R'
OLL CORNERS -yellow face	 Chameleon $\otimes$ 2 yellow; next to each other FUR'U'r' FRF'
 Sune $\otimes$ 1 yellow, rest clockwise RU'R'U RU2R'	 Bowtie $\otimes$ 2 yellow diagonal F'RU'R'U'r' FR

ADJACENT CORNER SWAP

exactly one face shows headlights

• T: 1 headlights; pairs B-left F-left
RUR'U' R'F R2UR'U' RUR'F'

Ja L: solid; pairs B-right F-left B-front
x R2FRF' RU2 r'Ur U2

Jb L: solid; pairs B-left F-right B-back
RUR'F' RUR'U' R'F R2UR'

Aa L: headlights; pairs F-right B-front
x L2D2 L'U'L D2 L'U'L'

Ab L: headlights; pairs B-right B-back
x' L2D2 LUL' D2 L'U'L

F: L solid
R'U'F' RUR'U' R'F R2U'R'U' RUR'UR'

One face shows headlights: two corners of that face match. Hold as

ADJACENT CORNER SWAP

continued

Ra L: headlights; pairs F-left
RUR'U' RUR DR'URD' R'U2UR'

Rb L: headlights; pairs B-left
R2F RURUR' FR U2RU2R

Ga: headlights; pairs F-right
R2UR'UR'U' RU'R2 U'D R'URD'

Gb L: headlights; pairs B-back
R'UR UD' R2UR'UR'U' RU'R2D

Gc L: headlights; pairs B-right
R2U'R'UR'U' R'UR2 UD' RU'R'D

Gd L: headlights; pairs F-front
RUR' U'D R2U'R'UR'U' UR'UR2D'

Learn in the printed order, three weeks about six weeks; Ja and Jb on the same day — one is the mirror of the other.

ONE-LOOK PUZZLE ADJACENT CORNER SWAP — exactly one face shows headlight	ADJACENT CORNER SWAP — continued
 • T 1. headlights; pairs R-Left RUR'U' R'F R2UR'U' RUR'F'	 Ra 1. headlights; pairs R-Left RU'R'U' RUR DR'URD' R'U2UR'
 Ja 1. solid; pairs R-Right F-Left R-front x R2FRF' RU2 r'Ur U2	 Rb 1. headlights; pairs R-Left R2F RURUR' F'R U2RU2R
 Jb 1. solid; pairs R-Left F-Right R-back RUR'F' RUR'U' R'F R2UR'	 Ga 1. headlights; pairs F-right R2UR'UR'U' RU'R2 U'D R'URD'
 Aa 1. headlights; pairs F-right R-front x L2D2 L'UL' D2 L'UL'	 Gb 1. headlights; pairs R-back R'UR' UD' R2UR'UR'U' RU'R2D
 Ab 1. solid; pairs R-right R-back x' L2D2 LUL' D2 LU'L	 Gc 1. headlights; pairs R-right R2UR'UR'U' RU'R2 UD' RU'R'D
 F 1. solid R'UF' RU'R'U' R'F R2UR'U' RU'RUR'	 Gd 1. headlights; pairs R-front RU'R' U'D R2UR'UR'U' UR'UR2D'
One face shows headlights: two corners of that face match. Hold as	

[illegible][illegible]

Print at 100% / Actual size — do NOT fit to page. Two-sided, flip on the SHORT edge. The nume

The numeral on both faces is your check: cut ONE card first. This bar must measure

exactly 80 mm:

■ 00 1/3-B FIRST SOLVE

[illegible]

One algorithm, up to three times: dot to hook to line to cross.

[illegible]





















The cube will look destroyed while you do this. Keep going – the
 impossible corner is now in the center. Never turn the whole cube.
 last top turn brings it back.

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UNLOCK CARD 2 – five scrambles in a row with this card face down. □ MASTER: those five under 3:00. My target: _____. □ NEXT: 2/3 TWO-LOOK

TWO-LOOK ■■ 2/3-B TWO-LOOK ■■ 2/3-B

PIPER P.F. P210PUL PULPES
T headlights on ONE face – hold that face LEFT
Yellow face: Sune, read again, repeat – at most 3 (see the badges).
Corners: T-Perm. No headlights to hold left? Run it twice.

Z VS H
 I no headings — the two swapping corners sit diagonally
 Both show headlights on all four faces. The edges between them

1. **100% DONE** — corners home, top and not sorted
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WORDS was reassembled wrong. Top it out and reveal it.

tree top turn that lines a case up

71-071 70 71-071 71-071 70 71-071

07W 70W 07W07W 0W 07W 70W 07W07W 0W

UNLOCK C/CAD 3 - fifteen last layers in a row, each finished in exactly two algorithms, case named out loud first. A repeat does not count. ☐ M4S1ER: solves averaging under 1:00. NEXT: 3/3 ONE-LOOK PLL

 • Y pairs F-left R-back
 • Ua B solid; F L R headlights; 3 edges counter-

• UB B solid: F-L B headlights: 3 edges clockwise

Na pairs B-left F-right L-front R-back

Nb parts: B-right F-left L-back R-front

HOW TO LOOK **NO R** **E** nothing matches

3. Line the case up as drawn, run it, turn the top again to finish —

PLT = slide the top pieces home - headlight = two matching
corners on one face - AUF = the free top turn that lines a case up -
week.

Next, the other 57 top-layer cases, and the first two layers solved as pairs – drilled with a cube in the app: cubepath-six-vercel.app/practice

THE LADDER
WORDS
1/3 FIRST SOLVE - 9 algorithms - unlock: five solves, card face
OLL = make the whole top face yellow - PLL = slide the top pieces

3/3 ONE-LOOK P.L. + 15 - done: all 21 named, twice on separate face - trigger = a short algorithm your hands run as one unit - AUF = the free top turn that lines a case up days.

WHY THERE ARE ONLY THREE

adjacent corner swap = the swapping corners share an edge .
diagonal corner swap = the swapping corners sit opposite
A card is good at closed set of cases you tell apart by sight . It is
bad at a full non-trivial case shuffling and look ahead stuff.

WHY THERE ARE ONLY THREE

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A card is good at closed set of cases you tell apart by sight . It is
bad at a full non-trivial case shuffling and look ahead stuff.

the app, with a real cube and randomised setup. This set stops where the medium stops.

Set V.10 - reprint any single card at cubepath-six.vercel.app/print - print at 100% / Actual size, never Fit to page.

Print at 100% / Actual size / NOT fit to page This sized fit on the SHOP! add The numerical on both foods is your absolute ONE word first This best must measurement

Downloaded from <http://ajphaphysiol.physiology.org/> at University of California, San Diego on September 11, 2015